



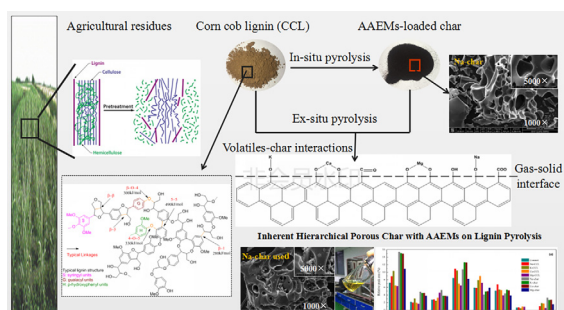
Influence of inherent hierarchical porous char with alkali and alkaline earth metallic species on lignin pyrolysis



Shaoqing Wang, Zhihe Li, Xueyuan Bai*, Weiming Yi, Peng Fu

School of Agricultural Engineering and Food Science, Shandong Research Center of Engineering and Technology for Clean Energy, Shandong University of Technology, Zibo 255000, China

GRAPHICAL ABSTRACT



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ABSTRACT

This study aimed to explore the influence of inherent hierarchical porous char with alkali and alkaline earth metallic species (AAEMs) during pyrolysis of lignin derived from agricultural crop residues in a laboratory fixed-bed at 550 °C. A catalytic strategy was implemented to investigate volatile-char interactions based on ex situ lignin pyrolysis. The physico-chemical properties of the AAEMs-loaded char were characterized by FTIR, XRD, SEM-EDX and N₂ nitrogen adsorption analyses. Results indicated that AAEMs-loaded char had a large specific surface area, hierarchical porosity, amorphous carbon structure, surface-active functional groups and highly dispersed metal species. Specifically, the specific surface area of AAEMs-loaded char was significantly reduced owing to coke deposition after interaction with pyrolysis vapours. Bio-oil composition revealed substantial increases in phenol, o-cresol, p-cresol and catechol. These increases were mainly attributed to demethylation, demethoxylation, or alkyl substitution reaction. The experimental results confirmed the occurrence of significant volatile-char interactions during lignin pyrolysis.

1. Introduction

Biomass as a carbon neutral energy has gradually become the ideal raw material for the preparation of bio-fuels and bio-based chemicals (Wang et al., 2017a). Specifically, agricultural residues have drawn attention as the future feedstocks for the synthesis of sustainable

bioenergy, which also provide potential application for the abundant agricultural crop residues in China (Ouyang et al., 2015). The components of agricultural crop residues mainly consist of cellulose, hemicellulose and lignin. Celluloses and hemicelluloses have been widely used to prepare cellulosic ethanol fuels or bio-gas from hydrolysis and fermentation, respectively (Liu et al., 2018b; Nayal et al., 2016).

* Corresponding author.

E-mail address: baixy@sdut.edu.cn (X. Bai).

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Notably, recalcitrant lignin is recognized as the largest barrier to biochemical transformation. In addition, the black liquor lignin from the paper industry remains with an annual output of 50 million tons, and about 95% of lignin by-products have been directly discharged into the rivers or subjected to combustion, leading to environmental concerns (Hu et al., 2017a). Therefore, effective pathways to disposal of these lignin-rich wastes are determined to boost bioeconomy and sustainability of industries.

Similarly, lignin is the second most abundant biopolymer after cellulose and the only non-petroleum renewable aromatic resource (Fan et al., 2017). Fast pyrolysis is currently one of the widely known techniques for lignin utilization, which can convert lignin to high-grade bio-oil as renewable feedstocks for the synthesis of fuels and chemicals (Liu et al., 2016). Previous studies on lignin pyrolysis have been conducted to understand the mechanism underlying pyrolysis (Guo et al., 2017; Wang et al., 2017b). In particular, the inherent alkali and alkaline earth metals (AAEMs) of lignin significantly affect its pyrolysis behavior. Zhang et al. (2017a) indicated that compared with woody biomass, agricultural residues contain more AAEMs, which can magnify the influence of pyrolysis products. Dalluge et al. (2017) demonstrated that organosolv corn stover lignin infused with AAEMs could affect the char yield and composition of bio-oil. Alkali metals (Li, Na, K and Cs), rather than alkaline earth metals (Mg, Ca, Ba and Cu(II)) exerted notable enhancing effects on char yield. Alkali metals (Li, Na, K and Cs) also promoted the conversion of the alkenyl side chains on volatile aromatics to light oxygenates. These interesting occurrences were attributed to AAEMs being able to alter dissociation of the internal linkage of lignin. Alkali metal additives were found to promote the decarboxylation reactions, decarbonylation reactions and removal of unsaturated alkyl side chain (Peng et al., 2014). More methoxyphenols were formed with Na_2CO_3 or K_2CO_3 . The NaOH and KOH additives promoted the deep deoxygenation of phenolic products, which caused a significant increase in the alkyl-phenols (“free” methoxy group). These studies showed that AAEMs could effectively influence the pyrolysis behavior of lignin.

A study found that pyrolysis reaction followed a reaction sequence from the outside to the inside, and internal pyrolysis vapors would inevitably pass through the outer surface of the char shell (Leijenhörst et al., 2016). In this process, the pyrolysis vapors were catalyzed by the metal species in the char shell. In the current study, the internal structure of the derived char could affect the dispersion of inherent AAEMs. On the other hand, their physico-chemical properties (i.e., specific surface area, pore structure, surface functional groups) determined the char catalytic reactivity to convert pyrolysis volatiles (Ancacouche et al., 2017; Kerkkaiwan et al., 2015; Oluoti et al., 2016; Wang et al., 2018; Yang et al., 2018). Recently, Song et al. (2014) reported on volatile-char interactions at low temperatures (400–600 °C) during tar destruction. Some O-containing species of tar were transformed to char and additional O-containing structures remained in the char matrix. Volatile-char interactions might have inhibited bio-oil production, particularly in fixed-bed reactors, as revealed in several relevant studies (Du et al., 2016; Kajitani et al., 2013; Zhang et al., 2017a). Feng et al. (2018) evaluated the effect of AAEMs pre/post-loaded biochar during gasification. The surface of AAEMs pre-loaded biochar exhibited a higher dispersion of K/Ca species and O-containing groups, which promoted the specific reactivity of biochar during gasification. Our previous research investigated the evolution of char structural characteristics and release properties of AAEMs during pyrolysis of corn straw (Fu et al., 2017). On the basis of previous studies and our related research, the effect of internal alkali metal of lignin derived from agricultural residues on the structural characteristics of char is not negligible and thus changes its physicochemical properties, thereby promoting the good dispersion of AAEMs on the char surface. However, studies are rarely reported on the effect of char coupling with AAEMs on the pyrolysis mechanism of lignin. Therefore, the aim of this study was to explore the evolution mechanism of AAEMs-loaded char

and volatile-char interactions.

In this study, AAEMs-loaded char was firstly prepared by in situ pyrolysis of AAEMs-impregnated lignin samples in a typical laboratory fixed-bed reactor. Ex-situ pyrolysis was then conducted to investigate the heterogeneous interactions between the AAEMs-loaded char and primary pyrolysis vapours. The AAEMs-loaded char was characterized by the Fourier transform infrared spectroscopy (FTIR), X-ray diffraction (XRD), scanning electron microscopy coupled with energy-dispersive X-ray spectroscopy (SEM-EDS) and N_2 nitrogen-adsorption. Bio-oil composition was detected by gas chromatography/mass spectrometry (GC-MS). By comparing the changes in char structures and bio-oil properties between in situ and ex situ pyrolysis, the evolution of AAEMs-loaded char structure and heterogeneous interactions between AAEMs-loaded char and primary vapors from the lignin matrix discussed in detailed. The influence of inherent char with AAEMs on lignin pyrolysis was clarified.

2. Materials and methods

2.1. Lignin collection and impregnated lignin sample with AAEMs

Lignin was extracted from a corn cob by using the Klason method and then labeled as corn cob lignin (CCL). Prior to impregnation, CCL was further purified by water-washing to minimize AAEMs content to serve as a blank control. About 5 mg of washed-CCL was added to 100 mL of 10 wt% AAEM (Na, K, Ca and Mg) salt solution. The mixed liquid was subsequently stirred at room temperature for 12 h. The uniform slurry was then washed with deionized water until neutral pH was reached, which would ensure the removal of excessive salt and retention of metal cations bound to the lignin matrix. Haddad et al. (2017) confirmed that AAEMs impregnation was followed with cation exchange mechanism. The impregnated CCL was transferred to the oven at 105 °C for 24 h. The prepared impregnated samples were denoted as Na-CCL, K-CCL, Ca-CCL and Mg-CCL.

2.2. Characterization of CCL samples

The ASTM D1102, ASTM E871 and ASTM E872 standard were applied in the determination of the ash, moisture, volatile and fixed carbon content. Ultimate analysis was conducted using an EA-3000 elemental analyzer (EuRo Vector, Milan, Italy). To ensure the analytical reliability of results, all runs were performed in triplicate; the reported averages had standard deviations below 1%. The respective data for the samples used in this study are provided in Table 1.

The AAEMs content was detected by inductively coupled plasma-optical emission spectrometry (ICP-OES) (Agilent 725ES). Prior to the experiments, the solid lignin samples were pretreated by microwave digestion in accordance with the US-EPA 3051 and 3052 methods. After digestion was completed, the supernatant was diluted to 100 mL in a

Table 1
Proximate and ultimate analyses of corn cob lignin used in this study.

Proximate Analysis (db ^a)	Analytical standard	Results-lignin
Moisture content, wt%	ASTM E871	6.08 ± 0.44 ^c
Volatile matter, wt%	ASTM E872	60.95 ± 0.52
Ash content, wt%	ASTM D1102	2.09 ± 0.01
Fixed carbon, wt%	By balance	30.89 ± 0.03
Ultimate Analysis (daf ^b)	Analytical instrument	Results-lignin
C, wt%	EuRo Vector, model CHNS/O 2400	61.28 ± 0.08
H, wt%		5.33 ± 0.01
N, wt%		0.90 ± 0.01
S, wt%		0.23 ± 0.02
O, wt%	By difference	32.27 ± 0.09

^a Dry Base.

^b dried and ash-free base.

^c the number after ± denotes Standard Deviation.

Table 2
Alkali and alkaline earth metal content of impregnated-lignin samples.

Sample	AAEM cation per kilogram lignin (mg/kg)			
	Na	K	Ca	Mg
Control (Pure Lignin)	578 ± 0.57	621 ± 1.07	235 ± 0.98	97.52 ± 0.54
Na-CCL	3156 ± 1.11	144.7 ± 0.58	142.5 ± 1.36	53.70 ± 0.78
K-CCL	185.2 ± 1.14	9654 ± 0.76	116.3 ± 2.49	46.65 ± 0.98
Ca-CCL	525.6 ± 0.13	546.3 ± 0.54	8482 ± 0.28	70.74 ± 0.38
Mg-CCL	495.6 ± 0.44	264.5 ± 1.12	196.6 ± 1.59	2422 ± 0.49

volumetric flask for subsequent analysis. It was then detected by ICP-OES to the AAEMs (Na, K, Ca and Mg) in lignin samples. The inorganic elements of different CCL samples are listed in Table 2.

Thermogravimetric analysis of AAEMs-loaded CCL samples was performed on an STA449C thermogravimetric analyzer (Netzsch, Selb, Germany). For each test sample, about 5 mg test material was loaded in a platinum crucible. During the experiment, the sample was heated at 10 °C/min from room temperature to 800 °C. High-purity nitrogen (99.999%) was supplied at a rate of 100 mL/min.

2.3. Pyrolysis experimental setup and procedure

A schematic of the pyrolysis experimental setup and the specific structure of the reactor is presented in Fig. 1. The device mainly consist of five units: the nitrogen supply unit, gas pretreatment unit, fixed-bed reactor unit, condenser unit and gas purifying unit. The internal diameter and height of the fixed-bed stainless steel reactor were 22 and 400 mm, respectively. The reactor was heated by three thermal insulation-type electric heating furnaces with a multi-stage temperature control. A high-density ceramic fiber material was chosen to maintain good heat preservation for each chamber. A char bed was added particularly to the bottom of the reactor tube to explore volatile-char interactions. The bottom of the reactor tube was designed with a Y-

shaped structure, which could effectively reduce the dead volume and backmixing effect of system to ensure volatiles propelled into the condensing system quickly. The condenser unit had two quartz glass tubes immersed in a low temperature bath that could keep the temperature below −10 °C by using an ethylene glycol coolant. To ensure the tightness of the device during the experiment, a mechanical switch was designed to control the dropping of a wire mesh cup.

The pyrolysis device was mainly used to accomplish two research objectives. The first aim was to investigate the effect of AAEMs on the thermal decomposition behavior of CCL individually. The pyrolyzed AAEMs-loaded char was then reserved after pyrolysis for subsequent use. For simplicity, the AAEMs-loaded chars were marked as Na-char, K-char, Ca-char and Mg-char. The second aim was to study the interaction mechanism between the primary pyrolysis volatiles of lignin and AAEMs-loaded char.

To pyrolyze the CCL samples individually, about 5 g of CCL sample was pre-loaded into the wire mesh cup. High-purity nitrogen (99.999%) was first heated to 300 °C with a preheater. The nitrogen then purged the reactor at a flow rate of 100 mL/min for 10 min. The reactor tube was subsequently heated to 550 °C at 10 °C/min. When the setting temperature was reached, the mechanical switch was turned to ensure that the cup entered the pyrolysis reaction zone along the guide rail. Pyrolysis time was set to about 45 min to ensure the complete release of CCL volatiles. The pyrolysis vapors ultimately entered the condenser with dichloromethane. Non-condensation gas passed via the purifying unit into the gas bag for subsequent analysis. After the reactor cooled to room temperature, the biochar was weighed. Simultaneously, char was collected for characterization analysis and used as catalyst to study the volatile-char interaction. The bio-oil was placed on a rotary evaporator at 40 °C to remove the dichloromethane for the analysis. Prior to distillation, about 2 mL of absorption solution was accurately weighed and then placed in a crucible, which was evaporated at 40 °C for 4 h, until a consistent quality was achieved. The remaining mass was considered in the concentration of bio-oil. For each pyrolysis experiment, the gas yield was followed the difference to assess mass balance. The volatile-char interactions experiment was similar to pyrolysis of the CCL

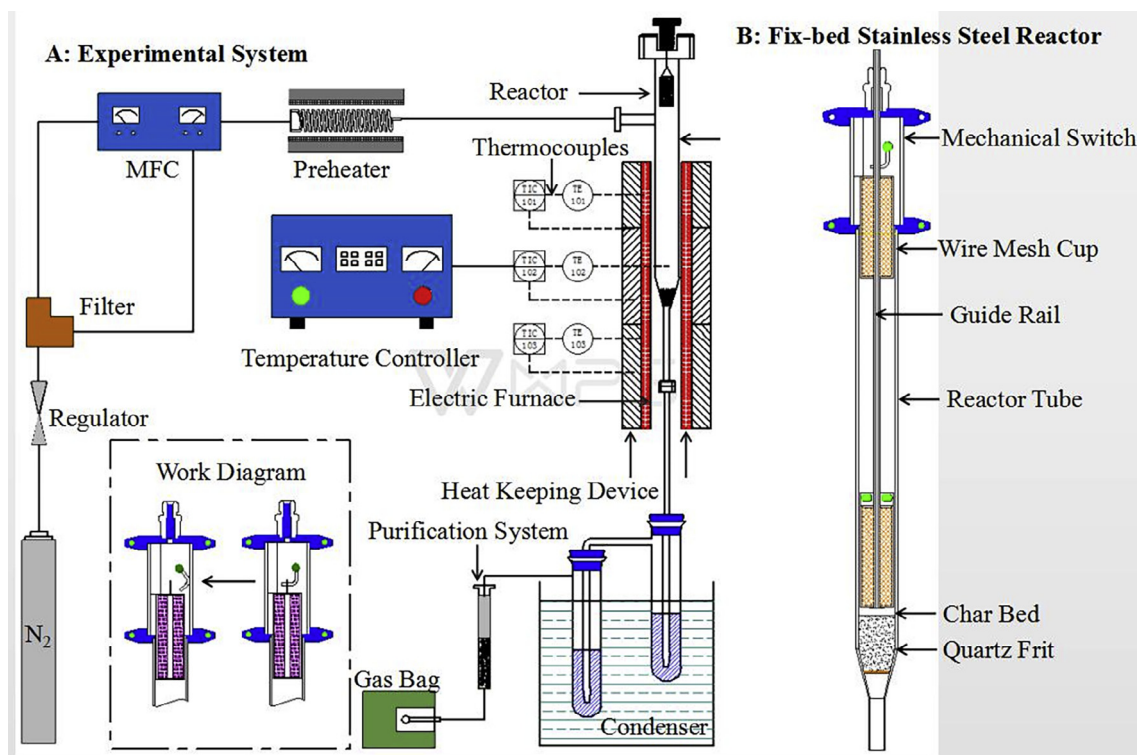


Fig. 1. Schematic of the pyrolysis experimental setup and specific structure of the reactor.

samples. Notably, the char should be pre-loaded on the quartz frit. The pyrolysis vapors of original lignin then passed through the char bed and underwent catalytic reforming. All experiments were repeated at least three times for the reproducibility of pyrolysis.

2.4. Char characteristics

FTIR (Nicolet 5700, Thermo Electron) was conducted to identify the evolution of functional groups during interaction. Prior to testing, the sample was thoroughly mixed with dry potassium bromide at a mass ratio of 1:100, and the fully ground mixture was pressed into a thin sheet with a diameter of about 13 mm under a pressure of 10 T/cm² for 1 min. Scanning was then performed in the spectral range of 4000–400 cm^{−1} at a spectral resolution of 2 cm^{−1}. Meanwhile, the pure KBr pellet of similar thickness was removed. The spectra obtained were removed from the pure KBr background, which is the infrared spectrum of a single sample.

An X-ray diffractometer (Bruker D8 Advance, Germany) equipped with Ni-filtered Cu and K α radiation (accelerating voltage: 40 kV, current: 40 mA) was used to record the diffraction patterns of different char samples. For testing, the scanning range was set to 10–60° with an angle step of 0.02°.

The morphology and surface elements of char samples were characterized by SEM (FEI Sirion 200) coupled with EDS (Oxford INCA), which was used to obtain a semi-quantitative analysis of the elemental distribution of the solid material microdomains. The samples were firstly coated with gold because of the absence of conductivity. Three scanning planes were selected for each experiment, and the average values represented the percentage of surface elements.

To characterize the surface texture properties of char samples, low-temperature nitrogen adsorption was performed on a Micromeritics ASAP 2460 multi-channel physisorption instrument at an operating temperature of −196 °C. Prior to each test, the sample was degassed at 100 °C vacuum atmosphere for 8 h to remove impurities and moisture. The Brunauer-Emmett-Teller (BET) method was employed to calculate the surface area. The pore volume and average pore diameters were calculated using the Barrett-Joyner-Halenda method.

2.5. Bio-oil characteristics

Bio-oil was detected by GC-MS (Agilent Technologies, 5973-6890N) with a capillary column DB-1701 (60 m $\times$ 0.25 mm $\times$ 0.25 μ m). Helium was maintained at a flow rate of 1 mL/min as the carrier gas. The split ratio and injection temperature were set to 1:60 and 280 °C, respectively. For each test, about 1 μ L of bio-oil was injected using 7683 series auto-injector. The column oven was programmed at an initial temperature of 40 °C for 2 min, heated to 240 °C at a heating rate of 5 °C/min, and held for 5 min. The operating parameters for the mass spectrometry were as follows: electron energy, 70 eV; scan mode range, 20–550 amu; ion source temperature, 150 °C; and interface temperature, 240 °C. The obtained mass spectra were tentatively identified by the National Institute of Standards and Technology mass spectral library to analyze the compounds in the bio-oil.

3. Results and discussion

3.1. Thermogravimetric analysis

To assess the effect of AAEMs on lignin thermal decomposition, the weight loss and weight loss rate curves of lignin and its AAEMs-loaded lignin samples are shown in Fig. 2. As observed, the lignin samples exhibited continuous weight loss during thermal decomposition, which is consistent with the findings by Ma et al. (2016). Pyrolysis was roughly divided into four stages. The first stage (25–150 °C) resulted from the dehydration of lignin samples with the low-temperature peak. Two small peaks were observed at the second stage (150–300 °C),

where contributed to abundant ferulic acid and p-coumaric acid units in the herbaceous lignin matrix (Liu et al., 2018a). Ester bonds existed in these units could cleave at a low temperature because of its lower bond dissociation energies. This occurrence was also proved by Hu et al. (2017b). Meanwhile, the pyrolysis behavior of other lignin extracted from woody biomass showed no significant weight loss at this stage (Liu et al., 2016). A broad shoulder peak was also observed in the AAEMs-loaded lignin samples rather than two small peaks in the original lignin samples, which indicated that the AAEMs inhibited the decomposition of the ferulic acid and p-coumaric acid units. The third stage (300–600 °C) was the most active pyrolysis process with about 40% weight loss. It was observed that the maximum peak temperature of AAEMs (Na, Ca and Mg)-loaded lignin samples was reached at a later time. K-lignin reached the peak weight loss earlier, but at a lower rate (2.63%/min) than those of other lignin samples. (E-Supplementary data for this work can be found in e-version of this paper online). These findings indicated that AAEMs suppressed the pyrolysis behavior of lignin samples at the third stage to a certain degree. The fourth stage (600–800 °C) corresponded to the deep cracking of lignin. Invisible weight loss of above 700 °C was observed. At the end of the pyrolysis process, char yield in the following increasing order was obtained: original-lignin K-lignin Na-lignin Ca-lignin Mg-lignin. This result was attributed to several factors: one was the new cross-linked structure (as detailed in the FTIR analysis) as the active sites to catalyze decomposition of lignin, which proved ionic strength effect of alkaline earth metal (Yang et al., 2006); another was that the AAEMs could promote the secondary cracking reaction of char residue.

3.2. Catalyst characterization

3.2.1. FTIR analysis of char

The spectrogram shows no significant change in the functional group type of the char. However, major changes are reflected in the intensity of the functional group. With the subsequent cation-exchange of AAEMs, the –OH group (peak 1) decreased significantly, accompanied by the formation of more –CO-AAEMs groups (Tang et al., 2017). The formation of a cross-linked structure in the presence of divalent alkali earth metal ions (Ca²⁺, Mg²⁺) is stronger than that by monovalent alkali metal ions (Na⁺, K⁺) with the formation of more hydrogen bonds (–O-AAEM–H–O–) confirmed by the increased depletion of –OH groups (peak 1), which is also consistent with the results of the study by Feng et al. (2018). The underlying reason was that O-containing groups can generate a negative charge on the surface, which can enhance adsorption of AAEMs cations in thermal reforming reactions. Uniformly, other organic O-containing groups (peaks 4, 7, 9 and 13) also exhibited a decreasing trend. The aliphatic –CH stretching vibration (peaks 1 and 2) was not visible after AAEMs impregnation, except for K-lignin. Zhang et al. (2016) indicated that potassium might be inserted into the edge of the carbon aromatic ring and caused the defects not only in the internal crystal lattice, but in the crystal layer spacing as well. This finding was consistent with XRD result. The new undulating peaks were related to the lone aryl CH wag (peak 9), two adjacent aryl CH wags (peak 10) and three or four aryl CH wags (peak 11) (Zhou et al., 2015). The aromatic C=C ring stretching (peaks 5 and 6) and aromatic C–H in-plane deformation slightly increased, which indicated the development of aromatization (Fu et al., 2017). In general, the functional groups on the surface of the char might provide more activity sites for the reforming volatile molecules (Wang et al., 2018).

3.2.2. XRD analysis

The crystallization characteristics of the internal structure of char can effectively reflect the cracking mechanism of the pyrolysis process (E-Supplementary data for this work can be found in e-version of this paper online). The XRD diffraction patterns of char samples exhibited broader diffuse diffraction peaks with more sharp and small diffraction

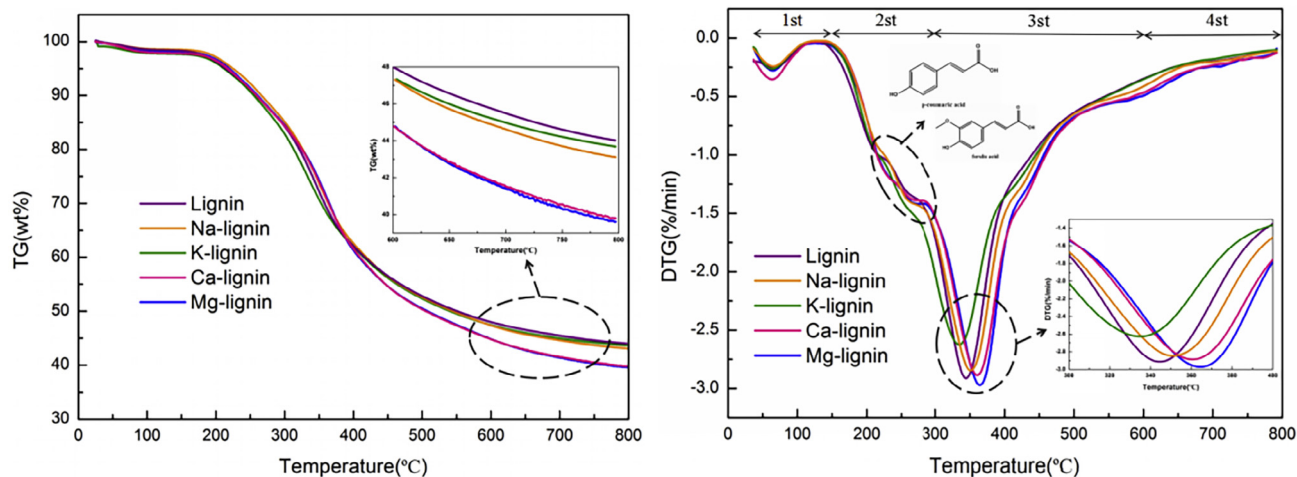


Fig. 2. Thermogravimetric curves for corn cob lignin and its AAEMs-loaded lignin samples.

peaks at 2 angles of 18–26, which corresponded to the C(0 0 2) peak for showing the stacking of monoatomic carbon layers in microcrystals, as reported by (Ahmad et al., 2018). The used AAEMs-loaded char exhibited a weaker background intensity with a reduction in amorphous carbon. This occurrence suggested that the volatiles could be reformed on catalytic active sites originated by metal-carbon complexes, which could result in the loss of numerous oxygen-containing functional groups (Volpe et al., 2017). This result was agreed with the FTIR analysis. Meanwhile, the graphitic might form carbonaceous precipitates on the surface of the char, as shown in the SEM micrographs. The aforementioned analysis indicates that the AAEMs-loaded char become more graphitic or stable after interaction. Another set of carbon-related (1 0 0) bands was associated with the graphite-like microcrystalline peaks at $2 \approx 44$, indicating an orderly stacking of aromatic layers (Fu et al., 2017). Notably, aromatic layers could enhance the chemical stability of char. However, the AAEMs-loaded char showed a slight reduction in ordering degree at (1 0 0) bands relative to that of lignin-char, suggesting inhibitory effect of inorganic elements on the condensation of aromatic rings (Gong et al., 2017). This result also proved that inorganic elements could affect the release of pyrolytic volatiles from lignin. A sharp peak at $2 \approx 31.76$ was clearly observed in the lignin-char, which could be explained by Na-containing crystalline peaks owing to the higher retention of sodium carbonate salt during the lignin preparation. The used K-loaded char exhibited a sharp peak at $2 \approx 28.48$, which belonged to K-containing crystalline peaks. This occurrence contributed to the large release of potassium during the volatile-char interactions (Kerckhaiwan et al., 2015). The EDX results showed the largest monovalent potassium content (4.62 wt%) on the surface of the char matrix.

3.2.3. Morphological analysis

The morphology of the char samples can be intuitively illustrated the microstructure changes before and after reforming process, which is conducive to support the volatile-char interactions (E-Supplementary data for this work can be found in e-version of this paper online). The pure lignin char exhibits a compact microscopic structure with fewer cavities and a macroporous hole. Char structure is mainly attributed to the collapse of lignin matrix caused by the rapid release of volatiles within the lignin, which is in accordance with the thermal ejection theory that lignin fragments are cracked and then expelled from the lignin matrix during lignin pyrolysis (Bayerbach and Meier, 2009; Lin et al., 2016; Lv et al., 2013). The surface structure of the AAEMs-loaded char becomes more irregular with the existence of large pores. This finding indicates that AAEMs can enhance degradation by destroying the aromatic structure of lignin, resulting in the release of a larger amount of volatiles (Wikberg et al., 2015). This result is consistent with

the TGA results. Meanwhile, AAEMs were more evenly distributed on the surface of the char from the EDS mapping (E-Supplementary data for this work can be found in e-version of this paper online). The figure shows that the oxygen element has a high degree of dispersion on the char surface. The strong electronegativity of the O (3.5) and C (2.5) exhibit strong binding with the weak electronegativity of the AAEMs [Na (0.9), K (0.8), Ca (1.0) and Mg (1.2)], which can form new metal-carbon complexes with redox properties (Okuno et al., 2005). The highly dispersed AAEMs species in the inherent hierarchical porous structure may be conducive for enhancing the char special catalytic activity, which would influence the reforming of lignin volatiles over char (Feng et al., 2018).

After the reforming of lignin volatiles, the structure of the AAEMs-loaded char changed significantly. A large number of small fragments or carbonaceous precipitates were distributed within the cavities of the AAEMs-loaded char because of the violent volatile-char interactions. Du et al. also proved the volatile-char interactions (Du et al., 2016). These fragments or carbonaceous precipitates might block the pore of the char, which was consistent with the decrease in surface area. From EDS results, the surface oxygen and AAEMs showed a certain increase, which indicated that the AAEMs species inside char matrix might migrate to the surface during thermal reforming (E-Supplementary data for this work can be found in e-version of this paper online). The aforementioned new metal-carbon complexes could act as a strong active adsorption site, which also increased the conversion of volatiles and also promoted the polycondensation of hydrocarbon volatiles to form coke (Tang et al., 2017; Feng et al., 2018).

3.2.4. Surface texture analysis

The specific surface area, pore volume and pore size of the char are listed in Table 3. The violent release of large amounts of volatiles created a well-developed micro-meso-macroporous structure during CCL pyrolysis. Fresh char catalysts mainly contained a larger micropore surface area (Char-control: 89.58 m²/g, Na-char: 169.28 m²/g, K-char: 54.98 m²/g, Ca-char: 188.99 m²/g and Mg-char: 236.06 m²/g). This indicated that the AAEMs could affect the pore structure of the char by changing the thermal behavior of CCL, causing a slight increase in external surface area derived from mesopores and macropores. Unexpectedly, K-char had a smaller surface area than the char-control. This finding was attributed to the large amount of potassium infused into the lignin matrix during the impregnation (see Table 2). Zhang et al. (2017a) also proved that corn cob was rich in potassium, which also suggested that potassium had a special binding ability with the corn cob matrix. The XRD and SEM-EDS analysis results indicate that the precipitation of potassium could exert an inhibitory effect on the development of the pore structure. The larger char surface area and

Table 3

Surface texture properties of AAEMs-loaded char sample during volatile-char interaction.

Sample	S _{BET} ^a (m ² /g)	S _{Micro} ^b (m ² /g)	S _{Ext} (m ² /g)	V _{tot} ^c (cm ³ /g)	V _{Micro} ^b (cm ³ /g)	V _{Ext} (cm ³ /g)
Char-control	106.09	89.58	16.51	0.0629	0.0460	0.0202
Na-char	192.48	169.28	23.20	0.0929	0.0796	0.0133
K-char	62.97	54.98	7.99	0.0310	0.0259	0.0050
Ca-char	212.31	188.99	23.32	0.0997	0.0888	0.0109
Mg-char	286.33	236.06	50.27	0.1396	0.1107	0.0289
Na-char-used	7.32	6.16	1.16	0.0149	0.0132	0.0017
K-char-used	4.64	4.21	0.43	0.0123	0.0105	0.0018
Ca-char-used	7.51	6.89	0.62	0.0141	0.0130	0.0011
Mg-char-used	8.22	6.72	1.50	0.0155	0.0125	0.0030

^a Determined by multi-point BET method.^b Calculated by t-plot method.^c Single point adsorption at P/P₀ = 0.993.

wide pore distribution could provide an excellent performance for the char-based catalyst. In addition, the AAEMs in the char matrix could improve its catalytic activity, as determined by Feng et al. (2018). However, the used char exhibited a significant reduction in the surface area and pore volume. These reductions were mainly due to the formation of new coke deposits during the interaction of volatiles with the porosity and AAEMs active site on the char matrix, which might cause blockage. Coke deposition indicated that the initial pyrolysis volatiles (particularly aromatic compounds) could easily form new coke on the surface of char during volatile-char interactions, as confirmed in several studies (Ancacouze et al., 2017; Wang et al., 2018; Yang et al., 2018).

3.3. Bio-oil analysis

3.3.1. Distributions of pyrolysis products for different feeds

The distributions of the pyrolysis products of different AAEMs-loaded CCL samples are shown in Fig. 3. The char and gas yields were enhanced with the infusion of AAEMs into the CCL matrix. In the current study, the thermogravimetric analysis results showed that the bond length and bond dissociation energy of lignin macromolecules were affected by the combination of lignin-containing oxygen functional groups with AAEMs, which could favor the release of benzene moieties in the CCL structure. In contrast to thermogravimetric analysis, the significant secondary reaction during the slow pyrolysis of CCL might cause deep cracking or repolymerization of the volatile with

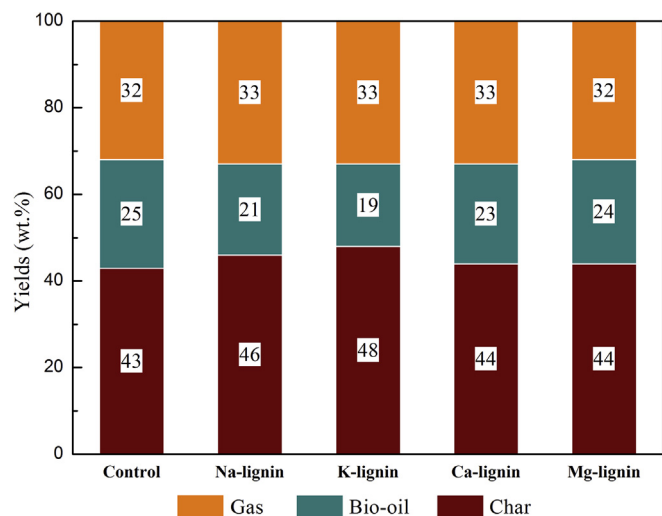


Fig. 3. Mass balance of pyrolysis products with different AAEMs-loaded CCL samples.

unsaturated side chains, which also showed consistency with the yield of bio-oil (Hosoya et al., 2008). Compared with the alkaline earth metals, the alkali metals produced more char, which was also attributed to the electropositivity of the metal itself. However, the alkaline earth metals markedly affected the gas yields, which was in agreement with a previous study by Liu et al. (2017). The intrinsic mechanism of change in pyrolysis products would be further discussed in the subsequent section.

3.3.2. Analysis of in situ pyrolysis behavior of impregnated AAEMs

The prepared bio-oil was mainly divided into four phenolic phases: P-type (phenol derivatives); G-type (guaiacol derivatives); S-type (syringol derivatives) and C-type (catechol derivatives). In addition, benzene derivatives and others (acids, furans, ketones, ether, esters, etc...) were also classified into two species in bio-oil, as illustrated in Table 4. For phenolic compounds, the alkali metals increased the phenol, o-cresol, 2-methoxyphenol contents with a newly produced catechol. However, the alkaline earth metals exhibited the opposite effect; it caused a slight increase in catechol. Notably, AAEMs promote the production of benzene derivatives. Hosoya et al. (2009) proposed the main pyrolysis pathways of methoxyphenol with demethylation or demethoxylation reactions and the radical-induced reaction. This occurrence could also be explained by the vulnerability of unstable phenolic compounds to secondary cracking and further recombination of free radical (Dorrestijn and Mulder, 1999; Yang et al., 2018). For example, 2-methoxyphenol could be directly transformed into saturated phenolic compound (phenol, o-cresol, and catechol) via demethylation or demethoxylation and further H-abstraction. The saturated alkyl-benzenes (1,2,4,5-tetramethylbenzene and 5-ethyl-1,2,4-trimethylbenzene) were formed followed by radical-induced reaction. The combination of the variation of these components in bio-oil suggests that the alkali metal could promote the secondary cracking of guaiacol and the alkaline earth metal with a slight effect. Simultaneously, 2-methoxyphenol exhibited a slight decrease with addition of the alkaline earth metal, which might repolymerize free functional groups to inhibit the formation of 2-methoxyphenol. However, the alkali metal could inhibit repolymerization, which ensured high 2-methoxyphenol content. However, 2-methoxyphenol also played a key role in the formation of char. Hosoya et al. (2009) selected a representative model compound of lignin for pyrolysis studies; the o-quinone methide derived from 2-methoxyphenol was proposed as a key intermediate for char formation. This occurrence could also explain the severe coke deposition problem during the catalytic pyrolysis of lignin (Mukkamala et al., 2012). Eom et al. (2012) found that the AAEMs could enhance C_γ-elimination reactions, which could form a vinyl ether structure. The vinyl ether structure was degraded into 2-methoxyphenol, 2-methoxy-4-vinylphenol, or condensed with other compounds to polymers, reducing amount of detectable monomeric compounds. A competitive relationship was clearly exhibited between C_γ-elimination reactions and the formation of the aforementioned o-quinone methide intermediates. Thus, such a competitive relationship should also be considered in the synthesis of special catalysts for the catalytic pyrolysis of lignin. Finally, we also detected a certain amount of 2,3-dihydrobenzofuran in the bio-oil with about 10 wt% in the bio-oil, which was mainly derived from the o-quinone methide intermediates. Previous research proposed the formation mechanism of 2,3-benzofuran by dissociated reaction, H-abstraction, and subsequent cyclization (Hosoya et al., 2009; Waser et al., 2017). In the present study, 2,3-dihydrobenzofuran could be formed from the hydrogen reduction reaction. 2,3-dihydrobenzofuran as a chemical product was mainly used as a flavor intermediate and a medicine intermediates (Yagunov et al., 2017), which further increased the value of lignin pyrolysis oil.

3.3.3. Analysis of ex situ pyrolysis behavior of AAEMs-loaded char

The release of lignin volatiles resulted in a change in char structure during in situ pyrolysis of lignin; however, the volatiles had no a

Table 4
Chemical compositions of the bio-oils produced from different CCL samples.

Abbreviation	RT.	Compound	Relative peak area (%)								
			Cont.	Na-CCL	Na-char	K-CCL	K-char	Ca-CCL	Ca-char	Mg-CCL	Mg-char
Phenol derivatives (P)-type			27.53	28.36	37.51	28.85	35.43	24.56	34.18	29.05	34.83
Ph	20.48	phenol	7.26	8.99	15.45	10.39	15.09	6.53	14.94	6.37	10.94
2MPh	21.90	2-methylphenol	2.21	2.00	4.89	3.05	4.53	1.94	4.54	1.66	3.80
4MPh	22.92	4-methylphenol	2.75	2.86	5.34	2.51	3.80	3.29	3.78	4.63	3.67
DMPH	24.25	2,5-dimethylphenol	1.25	1.09	1.78	1.23	1.63	1.40	1.65	1.47	1.81
4EPH	25.40	4-ethylphenol	9.98	10.81	7.07	7.71	7.59	8.27	6.53	11.66	9.99
4EMPh	26.58	4-ethyl-3-methylphenol	2.21	1.28	1.41	2.09	1.30	2.17	1.28	1.86	2.59
DEPh	28.29	3,5-diethylphenol	0.86	0.82	0.61	0.75	0.61	–	0.58	0.70	0.71
PP	28.91	4-(2-propenyl)phenol	1.01	0.51	0.96	1.12	0.88	0.96	0.88	0.70	1.32
Guaiacol derivatives (G)-type			25.38	30.78	21.90	27.27	25.34	27.08	25.34	28.61	22.42
MoPh	21.12	2-methoxyphenol	8.49	10.95	10.80	10.47	12.65	7.02	12.60	6.51	8.29
MoMPh	24.07	2-methoxy-4-methylphenol	4.43	4.57	3.70	3.79	4.07	4.71	4.00	6.23	3.77
EMoPh	26.40	4-ethyl-2-methoxyphenol	5.77	8.67	2.74	4.35	3.17	5.38	3.16	7.86	3.74
MoVPh	27.92	2-methoxy-4-vinylphenol	5.96	5.89	4.09	7.94	4.94	9.05	5.02	7.23	5.94
MoPPh	31.28	2-methoxy-4-(1-propenyl)phenol	0.73	0.70	0.57	0.72	0.51	0.92	0.56	0.78	0.68
Syringol derivatives (S)-type			5.77	7.51	4.40	5.96	5.38	6.26	5.26	4.82	3.90
DMoPh	29.30	2,6-dimethoxyphenol	5.11	6.79	4.40	5.43	5.38	5.32	5.26	3.92	3.90
DMoPPh	37.61	2,6-dimethoxy-4-(2-propenyl)phenol	0.66	0.72	–	0.53	–	0.94	–	0.90	–
Catechol derivatives (C)-type			0.00	1.03	3.29	1.87	2.66	1.74	2.73	0.57	1.57
o-BD	29.05	1,2-benzenediol	–	1.03	3.29	1.87	2.66	1.74	2.73	0.57	1.57
Benzene derivatives			4.11	8.4	8.1	7.7	7.12	6.45	6.98	7.15	7.69
Tol	8.75	Toluene	0.93	1.22	2.49	1.61	1.87	1.11	1.79	1.36	1.70
DMB	11.59	1,2-dimethylbenzene	–	–	0.76	–	0.54	–	0.55	–	0.52
MoMB	17.23	1-methoxy-4-methylbenzene	–	0.61	1.17	0.87	1.05	0.57	1.06	–	0.90
EMoB	20.07	1-ethyl-4-methoxybenzene	0.54	0.57	0.62	0.78	0.58	–	0.58	–	0.70
TMB	29.58	1,2,4,5-tetramethylbenzene	–	1.14	0.69	0.88	0.64	–	0.62	–	0.92
ETMB	30.62	5-ethyl-1,2,4-trimethylbenzene	–	0.95	1.23	1.60	1.20	1.57	1.12	1.19	1.80
TMoB	31.57	1,2,4-trimethoxybenzene	1.12	1.70	0.50	0.83	0.52	1.60	0.54	2.68	0.64
TMoMB	33.36	1,2,3-trimethoxy-5-methylbenzene	1.52	2.21	0.64	1.13	0.72	1.60	0.72	1.92	0.51
Others (Acids, Furans, Ketones, Ether and Esters)			17.37	13.55	11.09	15.24	11.64	17.25	11.09	16.14	17.52
AA	5.11	acetic acid	2.19	1.12	0.75	1.02	0.87	0.96	0.56	1.48	1.02
HBZD	19.33	2-hydroxybenzaldehyde	0.63	0.75	0.70	0.65	0.61	0.76	0.68	0.58	0.53
DHBF	27.75	2,3-dihydrobenzofuran	11.88	8.99	5.46	10.98	6.03	12.25	5.50	10.94	11.75
HMoBD	31.91	3-hydroxy-4-methoxybenzaldehyde	0.50	0.53	0.75	0.55	0.76	0.66	0.83	0.54	0.79
HPE	35.80	1-(3-hydroxyphenyl)ethanone	0.62	0.58	2.06	0.58	2.01	0.76	2.10	0.76	2.21
HDMoPE	39.79	1-(4-hydroxy-3,5-dimethoxyphenyl)ethanone	1.55	1.58	1.37	1.46	1.36	1.86	1.42	1.84	1.22
Unidentified			19.84	10.37	13.71	13.11	12.43	16.66	14.42	13.66	12.07

complete contact with the char matrix. By the aforementioned char characterization analysis, AAEMs-loaded char was found to have a large specific surface area, hierarchical porosity, amorphous carbon structure, surface-active functional groups or sites, and highly dispersed metal species. These advantages led to the excellent catalytic activity of AAEMs-loaded char in the volatile-char interactions. To further investigate these interactions during pyrolysis, a catalytic strategy was used to amplify this interaction. Fig. 4 and Table 4 show a substantial increase in phenol, o-cresol, p-cresol, and catechol, which were mainly attributed to the demethylation, demethoxylation or alkyl substitution reaction. In addition, the 2,3-dihydrobenzofuran content exhibited a significantly decreasing trend, which might have resulted from the cracking or reforming of the 2,3-dihydrobenzofuran over the AAEMs-loaded char (not including Mg-char) for the formation of phenol and ethane. Mg-char might exhibited a weak catalytic activity on 2,3-dihydrobenzofuran, almost retaining the 2,3-dihydrobenzofuran content. These findings indicated that compared with the other three metal-loaded char, Mg-char exhibited no significant catalytic effect. The syringol phenols contained two methoxy groups, which were easy to take off side chains and dehydration after the catalytic reforming of char (Eom et al., 2012). Consistent with the result for in situ pyrolysis, the AAEMs-loaded char played a crucial role in increasing the contents of benzene derivatives. The toluene content increased, compared with in situ pyrolysis. Unlike the control, there were more kinds of alkyl-substituted benzene and methoxybenzene. Meanwhile, bio-oil derived from the pyrolysis of alkaline earth metal-impregnated CCL had fewer types

of benzene derivatives, thus, the char itself exhibited a certain catalytic ability. This finding further showed that the alkaline earth metal on the surface of the char exerted a slight catalytic effect compared with the alkali metal. The experimental results also indicated that less acetic acid was detected after catalysis by AAEMs-loaded char; however, the 1-(3-hydroxyphenyl) ethanone content increased synchronously owing to the decarboxylation reaction (Zhang et al., 2017b). In general, the porous structure with AAEMs over char performed well during the volatile-char interactions. Compared with the alkaline earth metal, the alkali metals exerted a better effect on lignin pyrolytic char reactivity.

4. Conclusions

The evolution of the AAEMs-loaded char structure and the heterogeneous interactions between the AAEMs-loaded char and primary vapours from the lignin matrix was investigated. The obtained AAEMs-loaded char exhibited a better hierarchical porous structure with AAEMs over char, which exhibited good performance during the volatile-char interactions. Among the reasons was that the hierarchical porosity promoted the distribution of active sites, which subsequently improved the volatile-char interactions. Composition of bio-oil underwent major changes. Simultaneously, large amounts of coke deposition indicated the existence of the volatile-char interactions. This study can provide the basis to disposal lignin derived from agricultural residues.

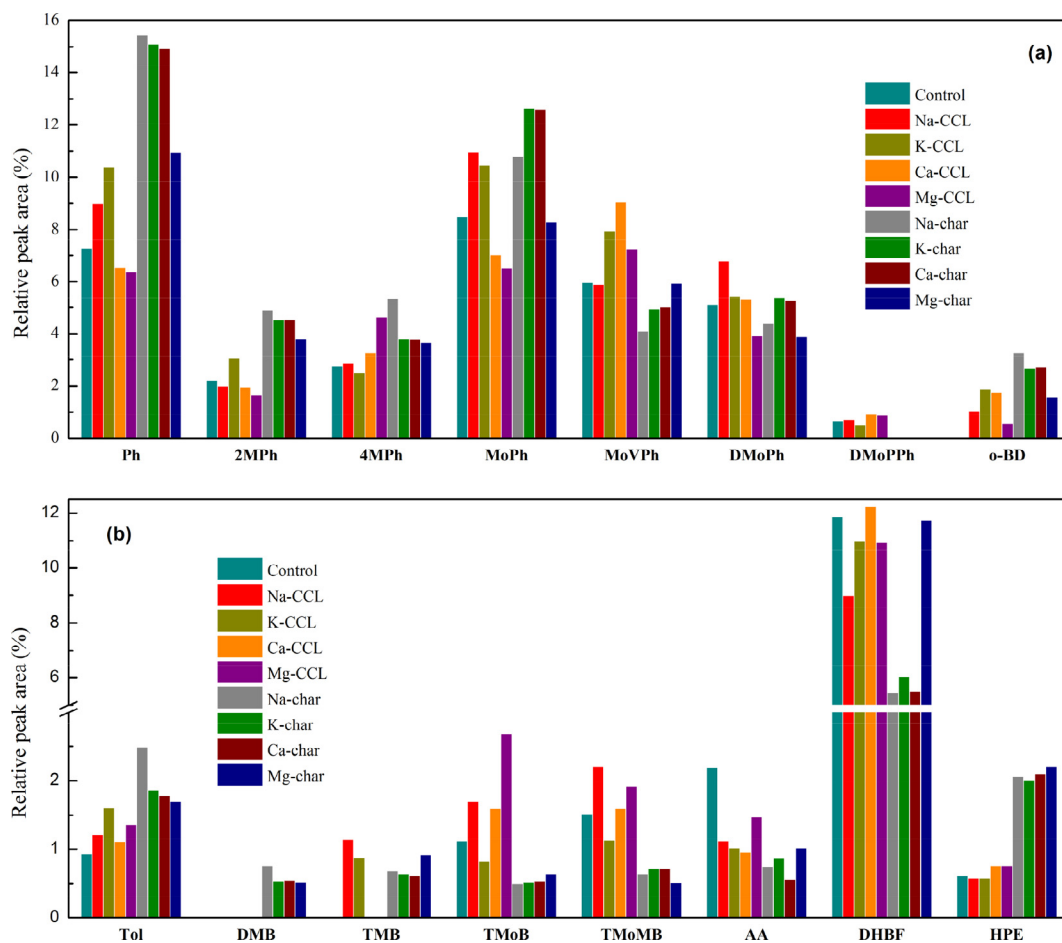


Fig. 4. Typical composition of bio-oil derived from pyrolysis of different lignin samples.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.biortech.2018.07.117>.

References

- Ahmad, J., Rashid, U., Patuzzi, F., Baratieri, M., Yun, T.Y., 2018. Synthesis of char-based acidic catalyst for methanolysis of waste cooking oil: an insight into a possible valorization pathway for the solid by-product of gasification. *Energy Convers. Manage.* 158, 186–192.
- Ancacouce, A., Dieguezalonso, A., Zobel, N., Berger, A., Kienzl, N., Behrendt, F., 2017. On the influence of heterogeneous secondary reactions during slow pyrolysis on char oxidation reactivity of woody biomass. *Energy Fuels* 31 (3), 2335–2344.
- Bayerbach, R., Meier, D., 2009. Characterization of the water-insoluble fraction from fast pyrolysis liquids (pyrolytic lignin). Part IV: structure elucidation of oligomeric molecules. *J. Anal. Appl. Pyrolysis* 85, 98–107.
- Dalluge, D.L., Kim, K.H., Brown, R.C., 2017. The influence of alkali and alkaline earth metals on char and volatile aromatics from fast pyrolysis of lignin. *J. Anal. Appl. Pyroly.* 127, 385–393.
- Dorrestijn, E., Mulder, P., 1999. The radical-induced decomposition of 2-methoxyphenol. *J. Chem. Soc. Perkin Trans. 4*, 777–780.
- Du, Z.Y., Wang, X., Zhang, Z.H., Feng, J., Qin, Y.H., Zhang, B.B., et al., 2016. Evolution properties of cellulose- and lignin-derived pyrolysis tars after interacting with coal chars. *J. Anal. Appl. Pyroly.* 122, 332–341.
- Eom, I.Y., Kim, J.Y., Kim, T.S., Lee, S.M., Choi, D., Choi, I.G., et al., 2012. Effect of essential inorganic metals on primary thermal degradation of lignocellulosic biomass. *Bioresour. Technol.* 104, 687–694.
- Fan, L., Zhang, Y., Liu, S., Zhou, N., Chen, P., Cheng, Y., et al., 2017. Bio-oil from fast pyrolysis of lignin: effects of process and upgrading parameters. *Bioresour. Technol.* 241, 1118–1126.
- Feng, D., Zhao, Y., Zhang, Y., Xu, H., Zhang, L., Sun, S., 2018. Catalytic mechanism of ion-exchanging alkali and alkaline earth metallic species on biochar reactivity during CO₂/H₂O gasification. *Fuel* 212, 523–532.
- Fu, P., Yi, W., Li, Z., Bai, X., Wang, L., 2017. Evolution of char structural features during fast pyrolysis of corn straw with solid heat carriers in a novel v-shaped down tube reactor. *Energy Conv. Manage.* 149, 570–578.
- Gong, X., Zhang, S., 2017. Changes in char structure due to inorganic matters during anthracite pyrolysis. *J. Anal. Appl. Pyroly.* 127, 170–175.
- Guo, D., Wu, S., Lyu, G., Guo, H., 2017. Effect of molecular weight on the pyrolysis characteristics of alkali lignin. *Fuel* 193, 45–53.
- Haddad, K., Jeguirim, M., Jellali, S., Guizani, C., Delmotte, L., Bennici, S., Limousy, L., 2017. Combined NMR structural characterization and thermogravimetric analyses for the assessment of the AAEM effect during lignocellulosic biomass pyrolysis. *Energy* 134, 10–23.
- Hosoya, T., Kawamoto, H., Saka, S., 2008. Secondary reactions of lignin-derived primary tar components. *J. Anal. Appl. Pyroly.* 83, 78–87.
- Hosoya, T., Kawamoto, H., Saka, S., 2009. Role of methoxyl group in char formation from lignin-related compounds. *J. Anal. Appl. Pyroly.* 84, 79–83.
- Hu, J., Zhang, Q., Lee, D.J., 2017a. Kraft lignin biorefinery: a proposal. *Bioresour. Technol.* 247, 1181–1183.
- Hu, J., Shen, D., Wu, S., Xiao, R., 2017b. Insight into the effect of ZnCl₂ on analytical pyrolysis behavior of cellulolytic enzyme corn stover lignin. *J. Anal. Appl. Pyroly.* 127, 444–450.
- Kajitani, S., Tay, H., Zhang, S., Li, C., 2013. Mechanisms and kinetic modelling of steam gasification of brown coal in the presence of volatile-char interactions. *Fuel* 103, 7–13.
- Krerkkaiwan, S., Mueangta, S., Thammarat, P., Jaisat, L., Kuchonthara, P., 2015. Catalytic biomass-derived tar decomposition using char from the co-pyrolysis of coal and giant leucaena wood biomass. *Energy Fuels* 29, 3119–3126.
- Leijenhorst, E.J., Wolters, W., van de Beld, L., Prins, W., 2016. Inorganic element transfer from biomass to fast pyrolysis oil: Review and experiments. *Fuel Process Technol.* 149, 96–111.
- Lin, H., Wang, S., Zhang, L., Ru, B., Zhou, J., Luo, Z., 2016. Structural evolution of chars

- from biomass components pyrolysis in a xenon lamp radiation reactor. *Chin. J. Chem. Eng.* 25, 232–237.
- Liu, C., Hu, J., Zhang, H., Xiao, R., 2016. Thermal conversion of lignin to phenols: Relevance between chemical structure and pyrolysis behaviors. *Fuel* 182, 864–870.
- Liu, C., Wang, X., Lin, F., Zhang, H., Xiao, R., 2018a. Structural elucidation of industrial bioethanol residual lignin from corn stalk: a potential source of vinyl phenolics. *Fuel Process. Technol.* 169, 50–57.
- Liu, G., Zhang, Q., Li, H., Qureshi, A.S., Zhang, J., Bao, X., et al., 2018b. Dry biorefining maximizes the potentials of simultaneous saccharification and co-fermentation for cellulosic ethanol production. *Biotechnol. Bioeng.* 115, 60–69.
- Liu, Z., Wang, L., Jenkins, B.M., Li, Y., Yi, W., Li, Z., 2017. Influence of alkali and alkaline earth metallic species on the phenolic species of pyrolysis oil. *Bioresources* 12, 1611–1623.
- Lv, G.J., Wu, S.B., Yang, G.H., Chen, J.C., Liu, Y., Kong, F.G., 2013. Comparative study of pyrolysis behaviors of corn stalk and its three components. *J. Anal. Appl. Pyrolysis* 104, 185–193.
- Ma, Z., Sun, Q., Ye, J., Yao, Q., Zhao, C., 2016. Study on the thermal degradation behaviors and kinetics of alkali lignin for production of phenolic-rich bio-oil using TGA-FTIR and Py-GC/MS. *J. Anal. Appl. Pyroly.* 117, 116–124.
- Mukkamala, S., Wheeler, M.C., Heiningen, A.R.P.V., Desisto, W.J., 2012. Formate-assisted fast pyrolysis of lignin. *Energy Fuels* 26, 1380–1384.
- Nayal, F.S., Mammadov, A., Ciliz, N., 2016. Environmental assessment of energy generation from agricultural and farm waste through anaerobic digestion. *J. Environ. Manage.* 184, 389–399.
- Okuno, T., Sonoyama, N., Hayashi, J.I., Li, C.Z., Sathe, C., Chiba, T., 2005. Primary release of alkali and alkaline earth metallic species during the pyrolysis of pulverized biomass. *Energy Fuels* 19, 2164–2171.
- Oluoti, K., Pettersson, A., Richards, T., 2016. Investigating the morphology and reactivity of chars from *Triplochiton scleroxylon* pyrolysed under varied conditions. *Bioresour. Technol.* 208, 94–99.
- Ouyang, W., Zhao, X., Tysklind, M., Hao, F., Wang, F., 2015. Optimisation of corn straw biochar treatment with catalytic pyrolysis in intensive agricultural area. *Ecol. Eng.* 84, 278–286.
- Peng, C., Zhang, G., Yue, J., Xu, G., 2014. Pyrolysis of lignin for phenols with alkaline additive. *Fuel Process. Technol.* 124, 212–221.
- Song, Y., Wang, Y., Hu, X., Hu, S., Xiang, J., Zhang, L., et al., 2014. Effects of volatile-char interactions on in situ, destruction of nascent tar during the pyrolysis and gasification of biomass. part I. roles of nascent char. *Fuel* 122, 60–66.
- Tang, H., Xu, J., Dai, Z., Zhang, L., Sun, Y., Liu, W., et al., 2017. Functional mechanism of inorganic sodium on the structure and reactivity of zhundong chars during pyrolysis. *Energy Fuels* 31, 10812–10821.
- Volpe, R., Messineo, A., Menendez, J.M.B., Millan, M., Reina, T.R., 2017. Evolution of chars during slow pyrolysis of citrus waste. *Fuel Process. Technol.* 158, 255–263.
- Wang, S., Dai, G., Yang, H., Luo, Z., 2017a. Lignocellulosic biomass pyrolysis mechanism: a state-of-the-art review. *Prog. Energy Combust. Sci.* 62, 33–86.
- Wang, S., Ru, B., Dai, G., Shi, Z., Zhou, J., Luo, Z., et al., 2017b. Mechanism study on the pyrolysis of a synthetic β -O-4 dimer as lignin model compound. *Proc. Combust. Inst.* 36, 2225–2233.
- Wang, S., Jiang, D., Cao, B., Hu, Y., Yuan, C., Wang, Q., et al., 2018. Study on the interaction effect of seaweed bio-coke and rice husk volatiles during co-pyrolysis. *J. Anal. Appl. Pyroly.* 132, 111–122.
- Waser, M., Roiser, L., Meisinger, N., Monkowius, U., Himmelsbach, M., Robiette, R., 2017. Asymmetric synthesis of 2,3-dihydrobenzofurans via a [4 + 1] annulation between ammonium ylides and in situ generated ortho-quinone methides. *Chemistry* 23, 5137–5142.
- Wikberg, H., Ohraaho, T., Pileidis, F., Titirici, M.M., 2015. Structural and morphological changes in kraft lignin during hydrothermal carbonization. *ACS Sustain. Chem. Eng.* 3, 2737–2745.
- Yagunov, S.E., Kholshin, S.V., Kandalintseva, N.V., Prosenko, A.E., 2017. 5-hydroxy-2,3-dihydrobenzofuran-derived polyfunctional antioxidants. *Russ. Chem. Bull.* 66, 1024–1029.
- Yang, C.Y., Lu, X.S., Lin, W.G., Yang, X.M., Yao, J.Z., 2006. TG-FTIR study on corn straw pyrolysis-influence of minerals. *Chem. Res. Chin. Univ.* 22, 524–532.
- Yang, Z., Lei, H., Zhang, Y., Qian, K., Villota, E., Qian, M., et al., 2018. Production of renewable alkyl-phenols from catalytic pyrolysis of douglas fir sawdust over biomass-derived activated carbons. *Appl. Energy* 220, 426–436.
- Zhang, H., Ma, Y., Shao, S., Xiao, R., 2017a. The effects of potassium on distributions of bio-oils obtained from fast pyrolysis of agricultural and forest biomass in a fluidized bed. *Appl. Energy* 208, 867–877.
- Zhang, J., Zhang, R., Bi, J., 2016. Effect of catalyst on coal char structure and its role in catalytic coal gasification. *Catal. Commun.* 79, 1–5.
- Zhang, Y., Cui, H., Yi, W., Song, F., Zhao, P., Wang, L., et al., 2017b. Highly effective decarboxylation of the carboxylic acids in fast pyrolysis oil of rice husk towards ketones using CaCO_3 as a recyclable agent. *Biomass Bioenergy* 102, 13–22.
- Zhou, S., Brown, R.C., Bai, X., 2015. The use of calcium hydroxide pretreatment to overcome agglomeration of technical lignin during fast pyrolysis. *Green Chem.* 17, 4748–4759.